

TECHNICAL PROPOSAL: PROJECT ORACLE

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1. Technical Executive Summary

The objective is to deploy a **Systematic Intelligence Layer** that bridges the gap between static charting (TradingView) and autonomous decision-making. We will implement a multi-modal AI architecture consisting of a Finance-focussed LLM for Pine Script generation and a Computer Vision agent for topological market analysis.

Key Performance Indicator: Latency from Market Event to AI-Signal Overlay:
< 250ms.

2. System Architecture

We adopt a microservices architecture to handle the high throughput of market data while maintaining low-latency AI inference.

```

    [Data Source: Binance/CME] --(WebSocket)→ [Ingestion Gate]
                                     |
    [User Input: Chat/View] --(gRPC)→ [DANTE ORACLE CORE]
                                     |   |
    [LLM Inference (PineScript)]   [Vision AI (Patterns)]
                                     |
[Charting UI: TradingView Lib] <---(Custom Datafeed API)--- [Signal Aggregator]

```

3. AI Model Engineering

3.1. Language Model: Pine-Coder-70B

We will fine-tune a Llama-3 based model specifically on the Pine Script v5 documentation and 50,000+ open-source indicators.

- **Objective:** Convert Natural Language into executable code.
- **Optimization:** Quantization (4-bit AWQ) to ensure sub-second token generation on NVIDIA L40S GPUs.

3.2. Computer Vision: Topological Pattern Recognition

Standard indicators rely on OHLCV math. Our model uses a **Convolutional Neural Network (CNN)** trained on chart screenshots to detect subjective patterns (Elliott Waves, Head-and-Shoulders) with a confidence interval overlay.

4. TradingView Integration

We will leverage the **TradingView Advanced Charts Library (JS)** for deep integration.

Component	Implementation Tech	Function
Custom Datafeed	UDF (Universal Data Feed)	Streaming real-time signals alongside price candles.
Overlay Layer	Canvas API / Pine Script v5	Visualizing AI confidence bands and predictive price paths.
Strategy Bridge	Webhooks / Broker API	One-click deployment of AI-generated strategies to live exchange.

5. Performance & Scalability

Backend Stack

- **Language:** Golang (Inflow/Outflow), Python (Inference).
- **Message Broker:** Apache Kafka for signal distribution.
- **Database:** ClickHouse (Time-series data for backtesting).
- **Orchestration:** Kubernetes (K8s) on AWS/GCP with GPU auto-scaling.

6. Resource Requirements & Timeline

Role	Quantity	Focus
ML Engineer	2	LLM Fine-tuning & Vision Model Training.
Fullstack Developer	2	TV Integration & React Side Panel UI.
Backend Engineer	1	High-frequency streaming pipeline (Go/Rust).

Implementation Phases

1. **Phase 1 (Wks 1-4):** Data Pipeline & Baseline Pine-Coder Training.
2. **Phase 2 (Wks 5-8):** Beta Integration with TV Advanced Charts.
3. **Phase 3 (Wks 9-12):** Vision Model Deployment & Stress Testing.